

Abstract

Among various plasma disruption mitigation techniques, massive material injection with cryogenic pellets has emerged as a leading method for controlling plasma behavior. Understanding the ablation of cryogenic pellets enables well-tuned mitigation. The Pellets at Columbia (PAC) test stand intends to provide direct ablation data under controlled high-energy conditions via an electron beam. Several PAC measurement systems and methods have been implemented that are required for effective pellet injection and ablation study. Such diagnostic methods include the microwave cavities, fast camera/lighting, pellet tracking code, and rocket effect pellet deflection angle analysis. Together, these diagnostics play an essential role in the investigation of cryogenic pellet injection, advancing the development of pellet technology. This poster will present the current status of these diagnostic systems.

Motivation

- PAC test stand is intended to collect direct ablation data under high energy particle flux
- Pellet Injection (PI) can be used as a fueling, disruption mitigation, or plasma control technique
- Direct pellet ablation has not been empirically studied in the fusion field, making PAC a unique and necessary test stand

Operation

- Injection chamber and ablation chamber are rough-vacuum while making and shooting pellets
- Cryogenic Hydrogen, Neon, and mixes species pellets are made and sheared from inner barrel with the force of a mechanical punch
- Pellets accelerate through the barrel into the ablation chamber to be ablated by the e-beam
- Diagnostic systems (camera, MCs) used to determine various traits of the pellets

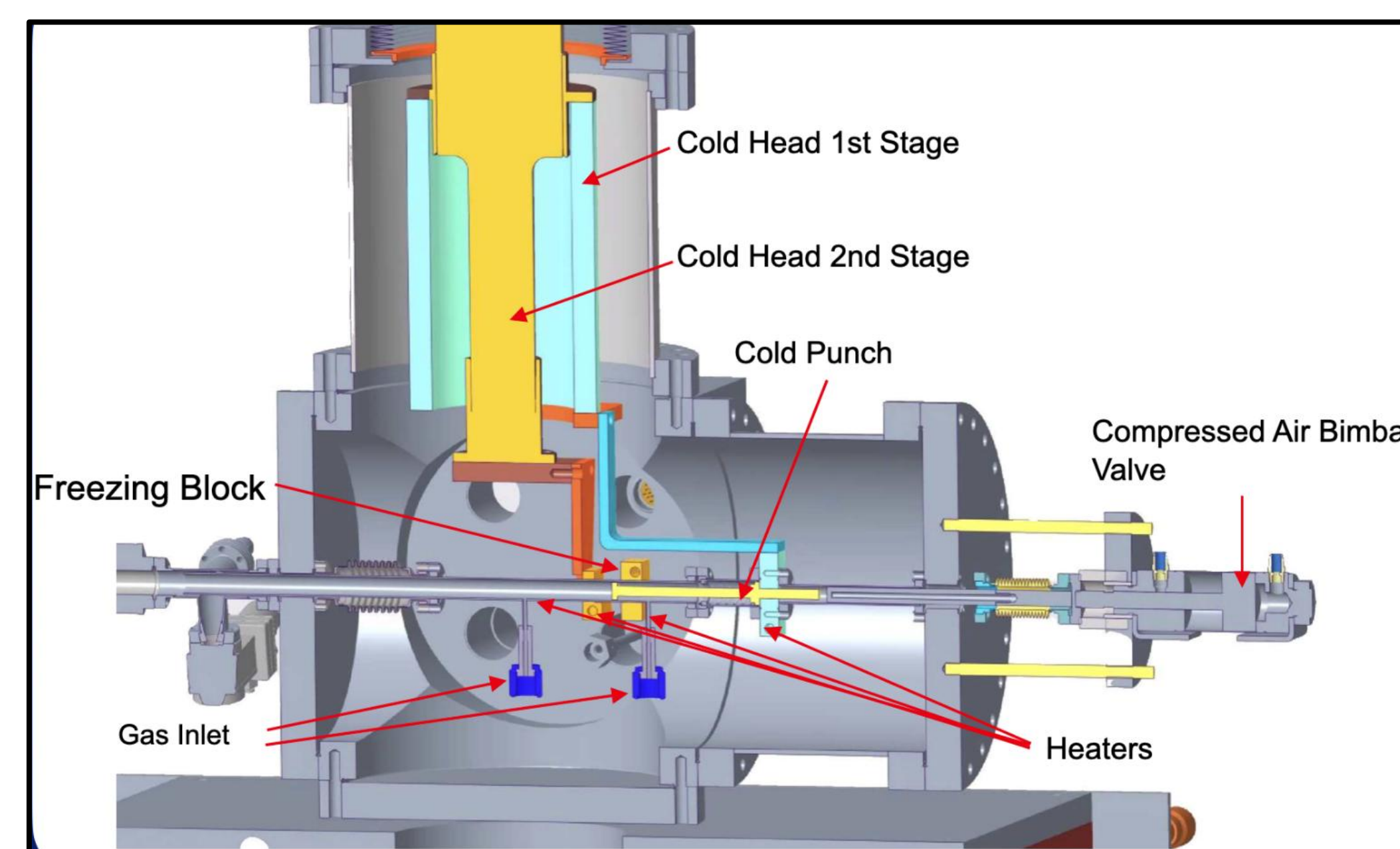


Fig.1. Labeled diagram of pellet injection chamber

Hardware

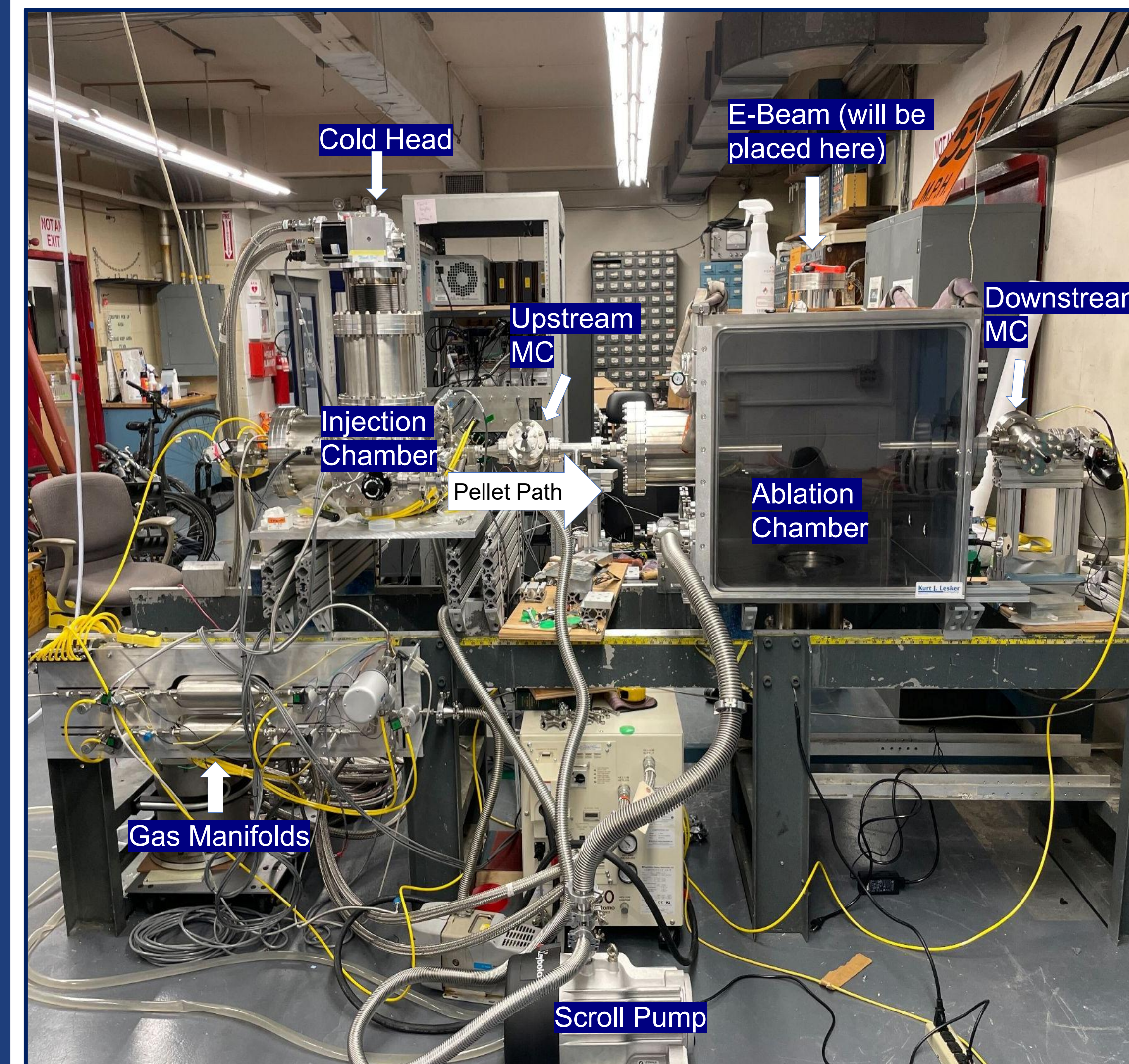


Fig 2. Labeled picture of the PAC hardware set-up including all relevant instruments and systems

Neon Pellets

- PAC's mechanical punch could buckle under the force needed to shear Neon pellets from the barrel

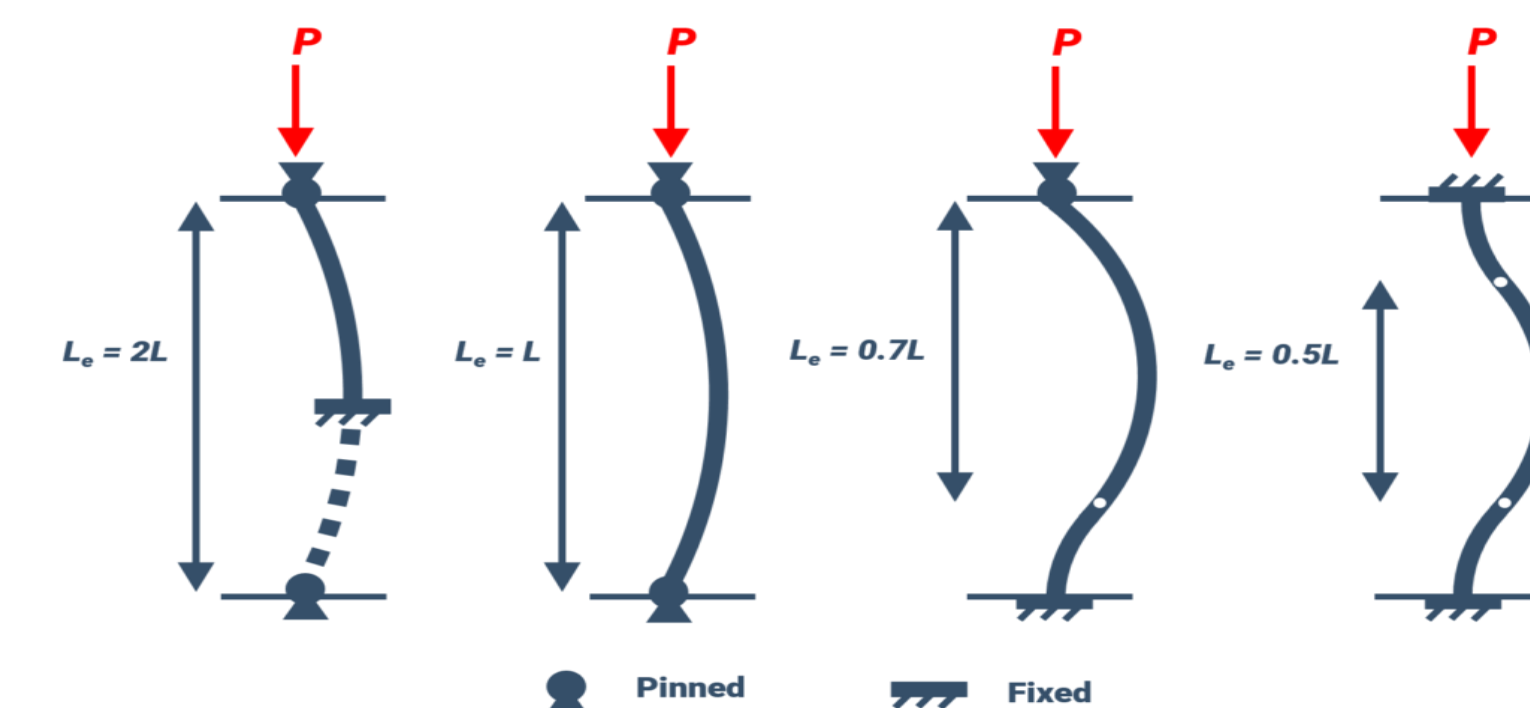


Fig. 3. Buckling force with different fixed and pinned configurations

- Using Eulers Buckling force equation below, the buckling force of the punch is ~250lbs

$$F_B = \frac{\pi^2 EI}{(KL)^2}$$

- Where E is the modulus of elasticity, I is the second moment of inertia (Ar^2), K is the buckling constant, and L is the effective length
- This force is then compared to the shear force of the pellet, calculated through the following equation

$$F_{shear} = P_{shear} * 2\pi rL$$

- Where the shear strength (P_{shear}) in MPa comes from the following relation [2]
- If shear force is greater than buckling force, the punch buckles
- Using shear and buckling forces, PAC was able to successfully inject Neon pellets without buckling the punch

Microwave Cavities

Microwave Cavities (MCs) are used to find the mass of PAC's cryogenic pellets

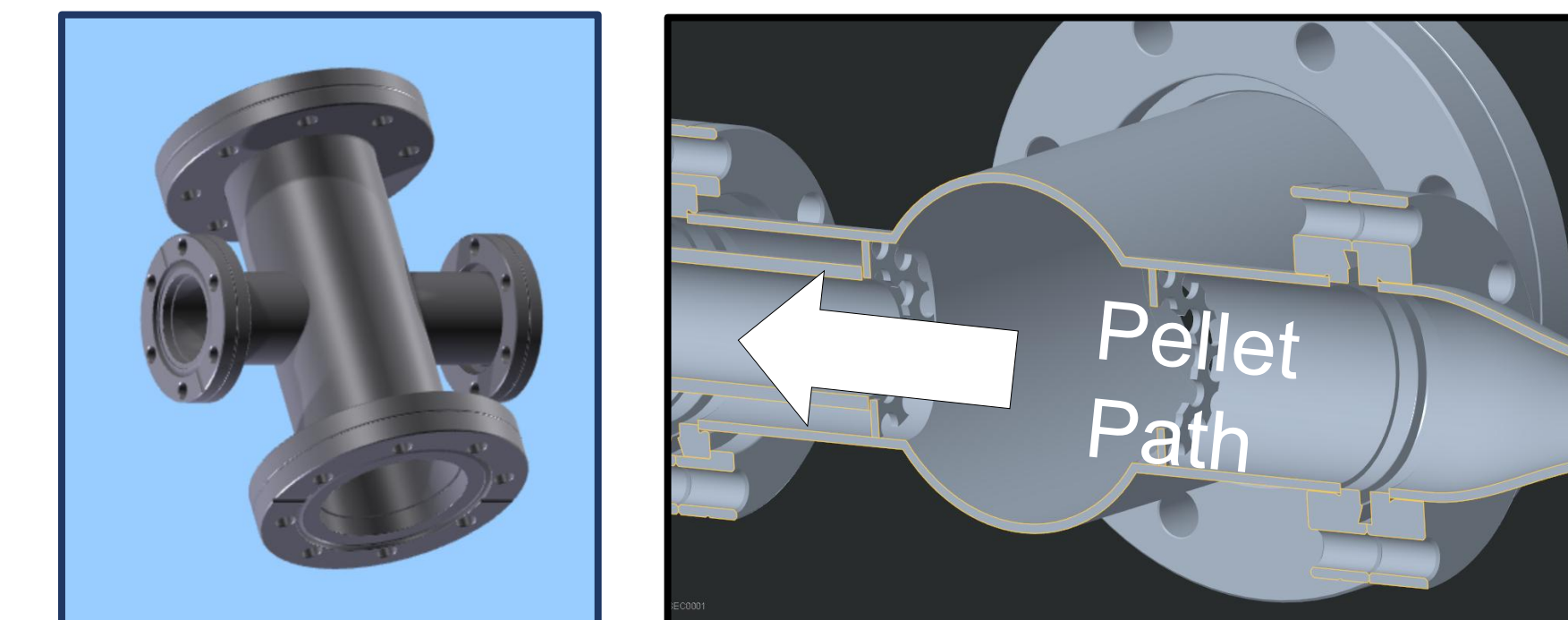


Fig. 3. Images of microwave cavities and the pellet path through them

- Peak voltage of the oscillator is found within the MCs
- Oscillator is tuned to ~60% of its peak voltage
- Pellets with a dielectric constant greater than the vacuum move through the cavities and MCs electric field changes
- Change in voltage on oscilloscope indicates a frequency shift in MCs, meaning a pellet has gone through

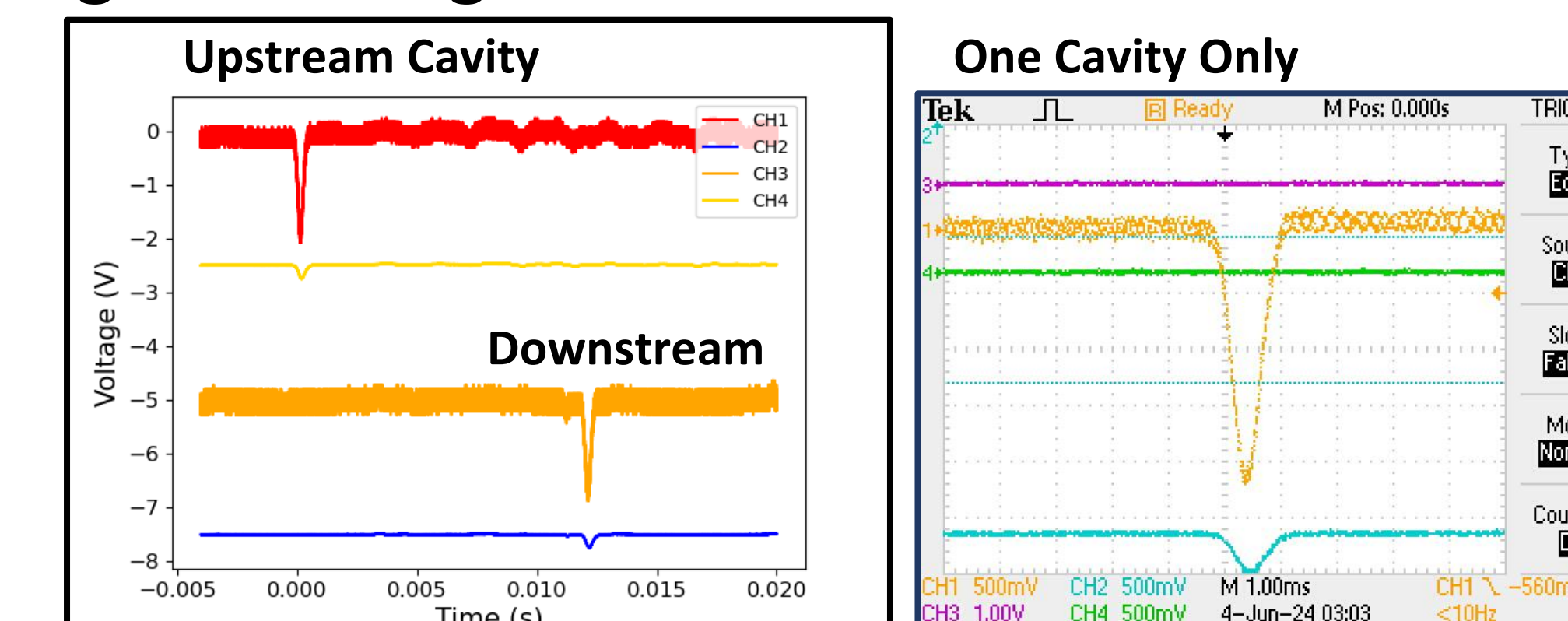


Fig 4. Plotted oscilloscope output of a pellet moving through the MCs. Left plot shows pellet moving through upstream and downstream MCs while right shows pellet moving through only one.

- Change in frequency within cavity is calculated by:

$$\Delta f_0 = \frac{\text{Pulse Out Voltage}}{\text{Cal Factor}} * \text{Detector Voltage}$$

- Pellet mass can be calculated by separating the pellet volume variable (below) and multiplying it by cryogenic density

$$\frac{\Delta f_0}{f_0} = \frac{-\left(\frac{3(e_r - 1)}{e_r + 2}\right)V_{\text{Pellet}}}{0.539 * V_{\text{Cavity}}}$$

- Where e_r is the cryogenic Hydrogen Dielectric Constant, V_{Pellet} is the volume of the pellet, V_{Cavity} is the volume of the MCs, and f_0 is the frequency inside the MCs
- Code script for PAC was created to analyze pellet masses using variables mentioned above
 - Creates a plot like figure four and finds the pellet mass/mass differences between MCs
 - Masses of pellets result in ~30 mg for Hydrogen and ~630 mg for Neon in MC 1

Pellet-Tracking

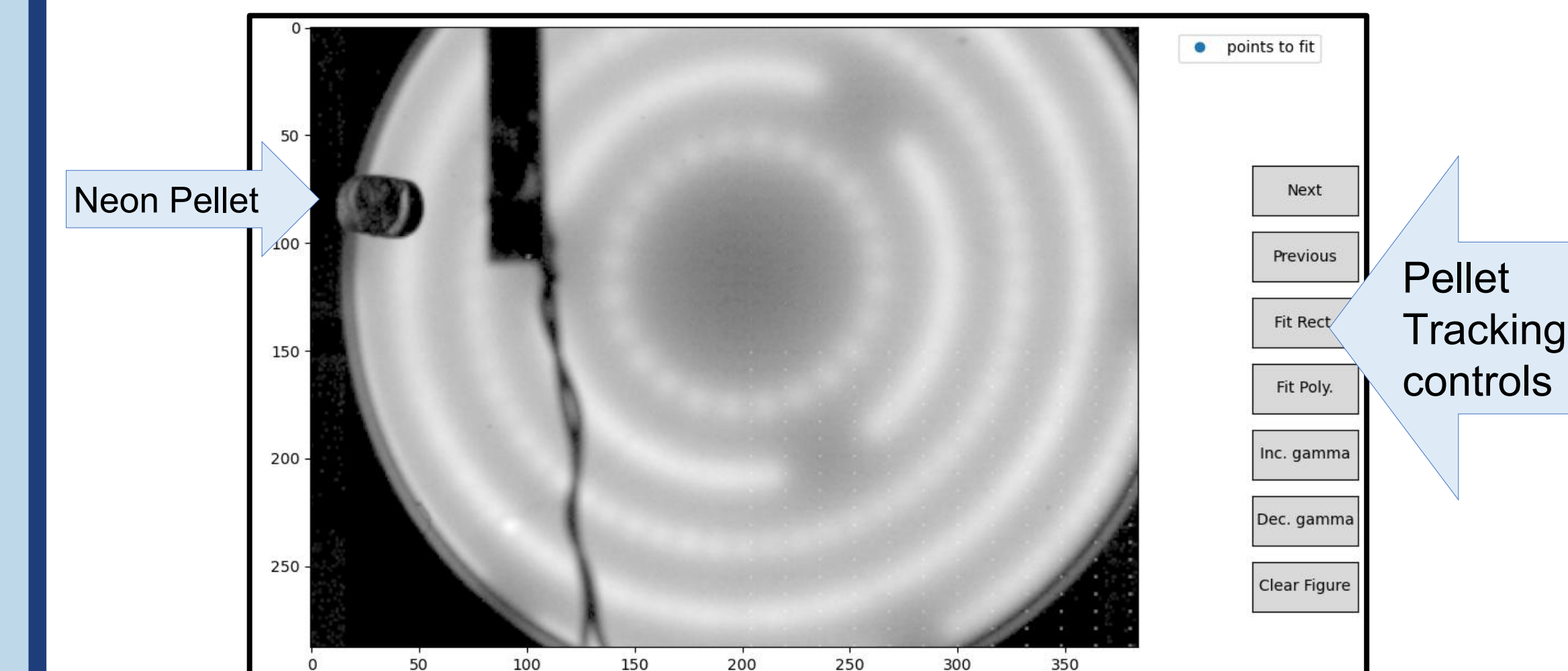


Fig. 5. Pellet tracking code interface with a Neon pellet located in the top left of the image

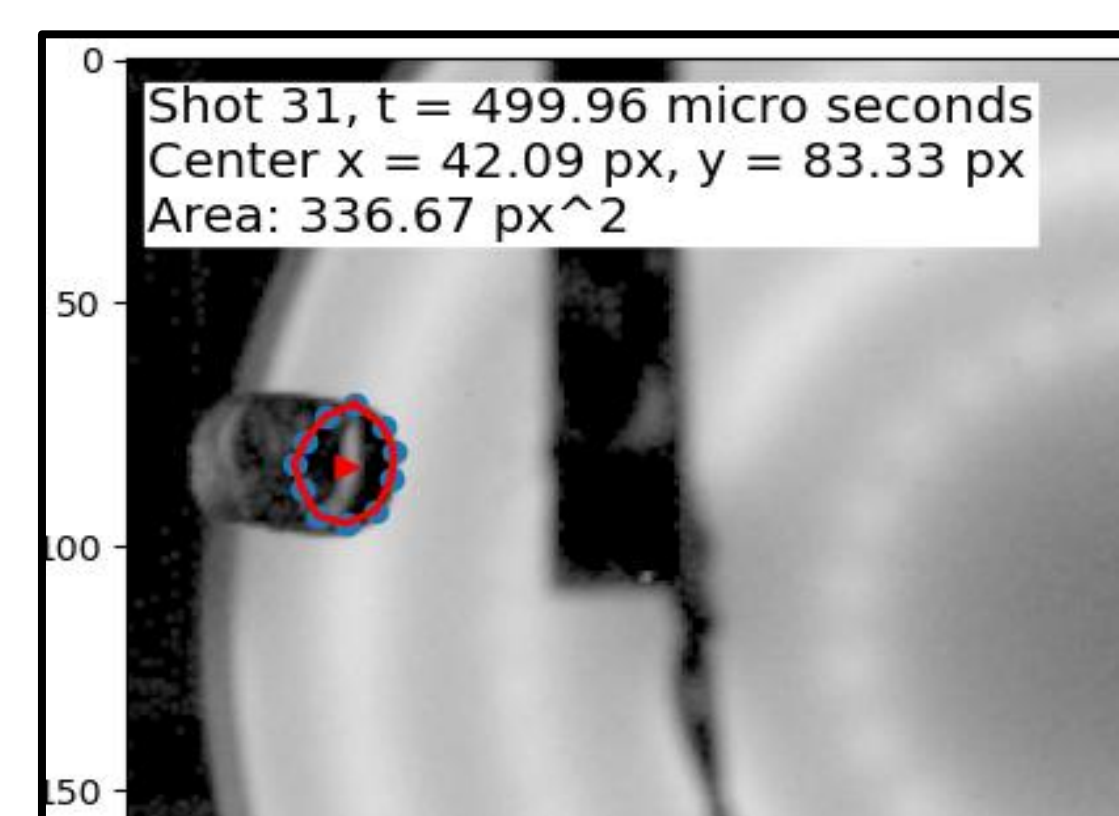


Fig. 6. Zoomed-in image of fitted pellet face with center details shown at the top

- Multiple points are plotted to outline a side of the pellet
- A polygon is fitted with the points
- The center of the pellet is computed
- To find the pellet angle of deflection:
 - Center of pellet is taken at first visible frame and frame right before ablation
 - Center of pellet is found the frame after ablation and in the very last visible frame
 - Change in angle before and after ablation is found
- Visual pellet tracking can determine how the pellets are ablated as they pass through e-beam
- Preliminary results suggest that PAC pellets have an initial angle of less than one degree before ablation

Future Work

- Measure multi-species pellet masses with MCs
- Use pellet tracking code to find angles of deflection due to pellet ablation
- Reduce standard deviation on mass differences between upstream and downstream MCs
- Obtain mass loss from ablation via MCs data

References

- [1]: Combs, S. K., and L. R. Baylor. "Pellet-Injector Technology—Brief History and Key Developments in the Last 25 Years." *Fusion Science and Technology*, 73, no. 4, 2018, 493–518, DOI: 10.1080/15361055.2017.1421367.
- [2]: Gebhart et al. (2021). Shear Strength and Release of Large Cryogenic Pellets from the Barrel of a Shattered Pellet Injector for Disruption Mitigation. *Fusion Science and Technology*, 77(7–8), 721–727. <https://doi.org/10.1080/15361055.2021.1874766>



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